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Math 400

Section 2.6 Limits at Infinity; Horizontal Asymptotes

Objective 1: Limits at Infinity and Horizontal Asymptotes.

1 Intuitive Definition of a Limit at Infinity Let f be a function defined on some interval (a, ∞). Then

$$\lim_{x \to \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

Example 1: Find the following limits.

(a.) $\lim_{x \to \infty} \frac{1}{x^2}$

2 Definition Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \to -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.

Example 1: (continued)

(b.) $\lim_{x \to -\infty} \frac{1}{x^2}$

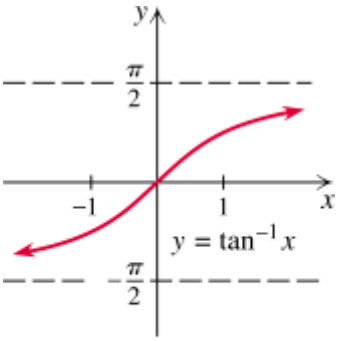
3 Definition The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \to \infty} f(x) = L \qquad \text{or} \qquad \lim_{x \to -\infty} f(x) = L$$

Example 2: Find the following limits.

$(a.)$ $\lim_{x \to \infty} \arctan x$

$(b.)$ $\lim_{x \to -\infty} \arctan x$



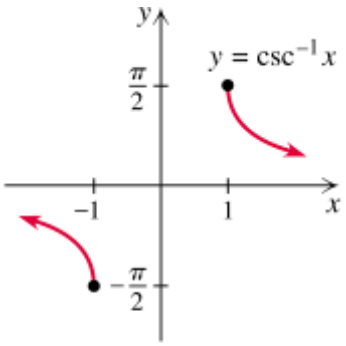
Domain $(-\infty, \infty)$

Range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Example 3: Find the following limits.

$(a.)$ $\lim_{x \to \infty} \csc^{-1} x$

$(b.)$ $\lim_{x \to -\infty} \csc^{-1} x$



Domain $(-\infty, -1] \cup [1, \infty)$

Range $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

Objective 2: Infinite Limits at Infinity.

Definition: Infinite Limits at Infinity

$\lim_{x\to\infty} f(x) = \infty$ says that $f(x)$ becomes arbitrarily large as x becomes arbitrarily large.

We can define the following limits in a similar manner:

$\lim_{x\to\infty} f(x) = -\infty$, $\lim_{x\to-\infty} f(x) = \infty$, $\lim_{x\to-\infty} f(x) = -\infty$

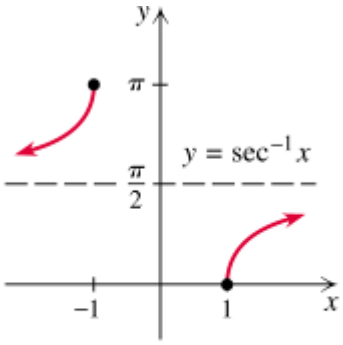
Example 4: Find the following limits.

$(a.)\ \lim_{x\to\infty} 2^x$

$(b.)\ \lim_{x\to-\infty} x^3 + x$

Example 5: Find the following limit. We will do this analytically.

$$\lim_{x\to\infty} \cos\left[\operatorname{arcsec}\left(e^{x}\right)\right]$$



Domain $(-\infty, -1] \cup [1, \infty)$

Range $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

Objective 3: Limits at Infinity for Rational Functions.

Example 6: Determine $\lim_{x\to\infty} f(x)$ for the following rational functions algebraically.

(a.) $\lim_{x\to\infty}\left(\dfrac{-9x^2-8x+11}{2x^2-10x+4}\right)$

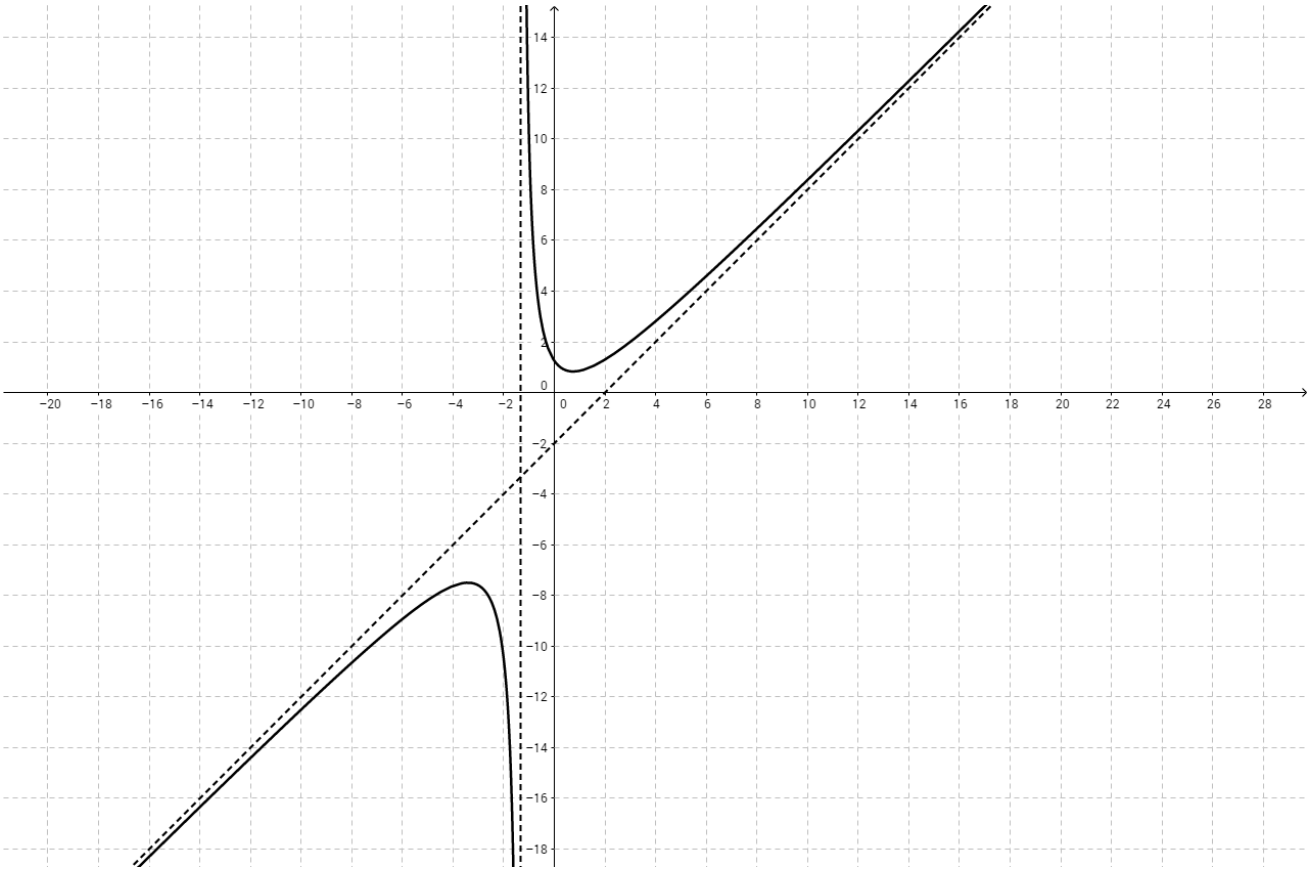
Example 6: (continued)

(b.) $\lim_{x\to\infty}\left(\frac{x^2-9x+1}{x^3+2x^2}\right)$

Example 6: (continued)

(c.) $\lim_{x\to\infty}\left(\frac{3x^2-2x+5}{3x+4}\right)$

Example 7: Let's look at the graph of $f(x) = \dfrac{3x^2 - 2x + 5}{3x + 4}$



The following is from Calculus 2nd Edition by Briggs and Cochran pages 93 – 94.

THEOREM 2.7 End Behavior and Asymptotes of Rational Functions

Suppose $f(x) = \dfrac{p(x)}{q(x)}$ is a rational function, where

$$p(x) = a_m x^m + a_{m-1}x^{m-1} + \cdots + a_2 x^2 + a_1 x + a_0 \quad \text{and}$$

$$q(x) = b_n x^n + b_{n-1}x^{n-1} + \cdots + b_2 x^2 + b_1 x + b_0,$$

with $a_m \neq 0$ and $b_n \neq 0$.

a. Degree of numerator less than degree of denominator If $m < n$, then $\lim\limits_{x \to \pm\infty} f(x) = 0$ and $y = 0$ is a horizontal asymptote of f.

b. Degree of numerator equals degree of denominator If $m = n$, then $\lim\limits_{x \to \pm\infty} f(x) = a_m/b_n$ and $y = a_m/b_n$ is a horizontal asymptote of f.

c. Degree of numerator greater than degree of denominator If $m > n$, then $\lim\limits_{x \to \pm\infty} f(x) = \infty$ or $-\infty$ and f has no horizontal asymptote.

d. Slant asymptote If $m = n + 1$, then $\lim\limits_{x \to \pm\infty} f(x) = \infty$ or $-\infty$, f has no horizontal asymptote, but f has a slant asymptote.

e. Vertical asymptotes Assuming that f is in reduced form (p and q share no common factors), vertical asymptotes occur at the zeros of q.

Objective 4: End behavior of algebraic functions.

Example 8: Determine the end behavior of the following function.

$$f(x) = \frac{2x^2 + 1}{\sqrt{x^6 - 4}}$$

Objective 5: End behavior of transcendental functions.

Example 9: Determine the end behavior of the following transcendental function by analyzing appropriate limits.

$f\left(x\right)=\ln(x-1)$

Example 10: Find the following limit.

$$\lim_{x\to\infty}\frac{e^{2x}-e^{x}}{e^{4x}+2e^{3x}}$$

Example 11: Find the following limit.

$$\lim_{x\to\infty} \frac{\cos x^2}{e^x}$$

Here is an extra example to look over before/while doing your homework.

Example 12: Determine the end behavior of the following functions.

$$f(x) = \frac{\sqrt[3]{x^6 - 10}}{-4x^2 + \sqrt{2x^4 - 3x}}$$

$$\lim_{x\to\infty} \frac{\sqrt[3]{x^6 - 10}}{-4x^2 + \sqrt{2x^4 - 3x}} = \lim_{x\to\infty} \frac{\sqrt[3]{x^6 - 10}}{-4x^2 + \sqrt{2x^4 - 3x}} \cdot \frac{\frac{1}{\sqrt[3]{x^6}}}{\frac{1}{\sqrt{x^4}}}$$

$$= \lim_{x\to\infty} \frac{\frac{\sqrt[3]{x^6 - 10}}{\sqrt[3]{x^6}}}{\frac{-4x^2 + \sqrt{2x^4 - 3x}}{\sqrt{x^4}}} = \lim_{x\to\infty} \frac{\sqrt[3]{\frac{x^6}{x^6} - \frac{10}{x^6}}}{\frac{-4x^2}{\sqrt{x^4}} + \frac{\sqrt{2x^4 - 3x}}{\sqrt{x^4}}}$$

$$= \lim_{x\to\infty} \frac{\sqrt[3]{1 - \frac{10}{x^6}}}{\frac{-4x^2}{x^2} + \sqrt{\frac{2x^4}{x^4} - \frac{3x}{x^4}}} = \lim_{x\to\infty} \frac{\sqrt[3]{1 - \frac{10}{x^6}}}{-4 + \sqrt{2 - \frac{3}{x^3}}} = \frac{\sqrt[3]{1 - 0}}{-4 + \sqrt{2 - 0}} = \frac{1}{-4 + \sqrt{2}}$$

$\Rightarrow y = \frac{1}{-4 + \sqrt{2}}$ is a horizontal asymptote

Example 12: (continued)

$$\lim_{x\to-\infty}\frac{\sqrt[3]{x^6-10}}{-4x^2+\sqrt{2x^4-3x}}=\lim_{x\to-\infty}\frac{\sqrt[3]{x^6-10}}{-4x^2+\sqrt{2x^4-3x}}\cdot\frac{\frac{1}{\sqrt[3]{x^6}}}{\frac{1}{\sqrt{x^4}}}$$

$$=\lim_{x\to-\infty}\frac{\frac{\sqrt[3]{x^6-10}}{\sqrt[3]{x^6}}}{\frac{-4x^2+\sqrt{2x^4-3x}}{\sqrt{x^4}}}=\lim_{x\to-\infty}\frac{\sqrt[3]{\frac{x^6}{x^6}-\frac{10}{x^6}}}{\frac{-4x^2}{\sqrt{x^4}}+\frac{\sqrt{2x^4-3x}}{\sqrt{x^4}}}$$

$$=\lim_{x\to-\infty}\frac{\sqrt[3]{1-\frac{10}{x^6}}}{\frac{-4x^2}{x^2}+\sqrt{\frac{2x^4}{x^4}-\frac{3x}{x^4}}}=\lim_{x\to-\infty}\frac{\sqrt[3]{1-\frac{10}{x^6}}}{-4+\sqrt{2-\frac{3}{x^3}}}=\frac{\sqrt[3]{1-0}}{-4+\sqrt{2-0}}=\frac{1}{-4+\sqrt{2}}$$

$\Rightarrow y=\dfrac{1}{-4+\sqrt{2}}$ is a horizontal asymptote